Launching the 2026 Autotrader

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LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- **Specify the decision pipeline.** Connect lagged data, EMA sentiment, single-index forecasts, positive utility preferences, target holdings, orders, fills, and ledger updates.
- **Enforce launch invariants.** Test information timing, numerical domains, self-financing, inventory, cash, execution, and risk limits before any external order is possible.
- **Stage an evidence-based launch.** Progress from deterministic simulation through walk-forward and paper trading to a capped pilot with monitoring, reconciliation, and a kill switch.

1 A Trading System Is More Than a Signal

An autotrader is a stateful control system coupled to market data, a broker, and an accounting ledger. The model proposes an action; the execution system determines what was actually filled; the ledger and risk engine determine what may happen next. A backtest that jumps directly from a signal to portfolio return omits the interfaces where many costly failures occur.

For the course prototype, a market signal comes from short- and long-horizon exponential moving averages (EMAs). With price observation P_t and span N, define

$$E_t^{(N)} = \alpha_N P_t + (1-\alpha_N)E_{t-1}^{(N)}, \qquad \alpha_N := \frac{2}{N+1}. \tag{1}$$

The initialization rule for $E_0^{(N)}$ must be declared because it affects early signals. A dimensionless crossover score available after the close of t is

$$z_t := \frac{E_t^{(s)}}{E_t^{(\ell)}} - 1, \qquad s < \ell, \tag{2}$$

where s and ℓ are the short and long spans. The notebook uses $-Gz_t$ to tilt allocation, where $G > 0$ is a gain. Because this quantity may be negative, it is better called a *sentiment or exposure tilt* than classical risk aversion, which is normally nonnegative.

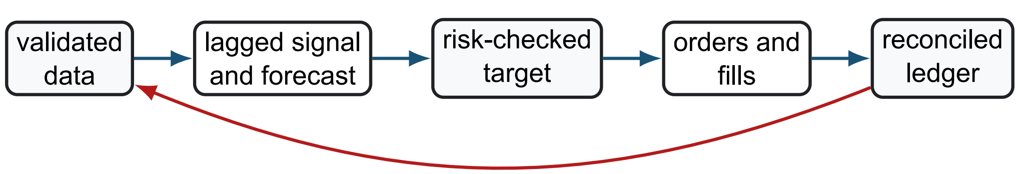

Figure 1: The minimum auditable autotrader loop. Each arrow is an interface with timing, validation, failure, and recovery requirements.

If the signal uses the close at date t, the first routine execution is at a price after that close, such as the next session's open. Same-close execution is valid only when the signal was computable before the order deadline from information already observed. Timestamp every observation, feature, decision, order, acknowledgement, fill, and ledger event.

2 Forecasts and Preferences Must Stay in Domain

For ticker i, a single-index forecast can be written

$$\hat{g}_{i,t} = \hat{\alpha}_{i,t} + \hat{\beta}_{i,t}\hat{g}_{m,t}, \tag{3}$$

where the estimates and market forecast use data available at the decision time. A separate transformation converts this forecast and the sentiment score into a strictly positive allocation preference $\gamma_{i,t} > 0$. One stable example is

$$\gamma_{i,t} := \gamma_{\min} + \log\left(1 + \exp\left[a\hat{g}_{i,t} - b\rho_{i,t} + cz_t\right]\right), \tag{4}$$

where $\gamma_{\min} > 0$ is a preference floor, a, b, c are declared scaling coefficients, and $\rho_{i,t} \geq 0$ is a positive risk measure such as forecast volatility. Standardize the inputs or bound the score to avoid overflow and unstable concentration.

This construction separates expected reward from risk. A SIM beta $\hat{\beta}_i$ is signed market exposure, not a universally positive risk scale. The inherited prototype computes $\hat{\beta}_i^{\lambda_t}$ for a generally non-integer, signed tilt λ_t; this is not real-valued when $\hat{\beta}_i < 0$ and is undefined at zero for some exponents. Use a positive ρ_i, or explicitly transform magnitude and sign separately.

Likewise, $\tanh(\hat{g}_i)$ lies in $(-1, 1)$ and can produce negative values. Those values cannot be inserted as exponents into the positive Cobb–Douglas solution derived in L14a. Equation Eq. 4, explicit exclusion of non-preferred assets, or a different constrained objective fixes the domain mismatch. A tiny forced holding of every negative-score asset is a policy choice, not the analytical optimum.

With positive preferences, normalized target weights are

$$w^*_{i,t} = \frac{\gamma_{i,t}}{\sum_{j\in\mathcal{S}_t}\gamma_{j,t}}, \qquad n^*_{i,t} = \frac{w^*_{i,t}B_t}{p^{\text{ref}}_{i,t}}, \tag{5}$$

before implementability constraints. Here $\mathcal{S}_t$ is the point-in-time eligible universe, B_t is deployable capital, and $p^{\text{ref}}_{i,t} > 0$ is a reference price used only to form targets. The execution engine must recalculate feasibility using actual order sides and fills.

Remark 2.1. Synthetic realism is a test claim, not an input label

An HMM-driven market factor combined with SIM residuals can generate useful controlled scenarios, but calibration to historical data does not make simulated paths realistic by declaration. Check regime duration, cross-sectional covariance, tails, volatility clustering, jumps, and drawdowns, and stress features the generator does not reproduce.

COMPANION NOTEBOOK

L15b: Synthetic Trade Bot →

Generate synthetic HMM/SIM prices, form EMA crossover signals, allocate shares, and compare the prototype with SPY and STRIPS benchmarks.

3 The Ledger Is the Source of Truth

Let $h_{i,t}^{-}$ and c_t^{-} be broker-confirmed shares and cash before an execution cycle. If signed fill $f_{i,t}$ is positive for a purchase and negative for a sale, post-fill state is

$$h_{i,t}^{+} = h_{i,t}^{-} + f_{i,t}, \tag{6}$$
$$c_t^{+} = c_t^{-} - \sum_i p_{i,t}^{\text{fill}} f_{i,t} - C_t(\mathbf{f}_t), \tag{7}$$

where $p_{i,t}^{\text{fill}} > 0$ and total execution cost $C_t \geq 0$ includes commissions, fees, spread, and modeled impact. Marked wealth at a declared liquidation convention $p_{i,t}^{\text{mark}}$ is

$$W_t = c_t^{+} + \sum_i h_{i,t}^{+} p_{i,t}^{\text{mark}}. \tag{8}$$

Order submission does not change the ledger; fills do. Partial fills, rejects, cancellations, and duplicate messages must be handled explicitly.

Theorem 3.1. Execution conservation check

Ignoring external deposits, withdrawals, dividends, and mark changes during the fill event, Eq. 6 implies

$$\left(c_t^{+} + \sum_i h_{i,t}^{+} p_{i,t}^{\text{fill}}\right) - \left(c_t^{-} + \sum_i h_{i,t}^{-} p_{i,t}^{\text{fill}}\right) = -C_t(\mathbf{f}_t). \tag{9}$$

Any larger discrepancy signals a sign, multiplier, quantity, price, fee, or event-processing error and should stop the strategy before another order cycle.

The notebook's pedagogical rule liquidates at a daily price and buys at a random perturbation of another price. For research, fill simulation should be side-aware and feasible: buys at or above the ask, sells at or below the bid, participation limited by volume, and fills bounded to positive prices. A Gaussian percentage perturbation has unbounded support and does not by itself model spreads or market impact.

Every cycle should assert invariants before and after trading:

- all prices, quantities, timestamps, and identifiers are valid and finite;
- the universe, corporate actions, and market session are current; stale or missing data cause no trade;
- cash, shorting, leverage, per-name weight, gross/net exposure, and turnover obey configured limits;
- order IDs are idempotent, open orders are included in risk, and broker state reconciles to the internal ledger; and
- a breached drawdown, connectivity, data-quality, or reconciliation threshold cancels open orders and disables new ones.

4 Launch Through Evidence Gates

The 2026 Autotrader should move through progressively more realistic environments. Advancement is conditional on predeclared evidence, not on one attractive wealth chart.

Unit and property tests cover EMA initialization, strict lags, preference-domain checks, allocation constraints, rounding, fill signs, conservation Eq. 9, duplicate events, and deterministic seeds. **Simulation** adds regime, gap,

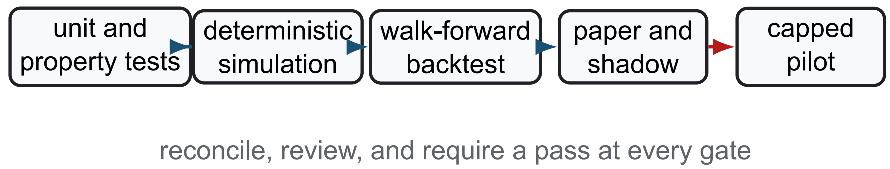

Figure 2: A safe launch ladder. Live capital appears only in the final capped pilot, after the identical decision and ledger code has passed offline and broker-connected tests.

spread, missing-data, rejection, partial-fill, and disconnect scenarios. **Walk-forward testing** repeatedly freezes parameters, trades a later interval, and reports every trial rather than selecting the best seed or window.

Paper trading exercises real market-data and broker interfaces without economic fills. A parallel shadow ledger compares intended orders, broker acknowledgements, hypothetical fills, and end-of-day positions. Only after sustained reconciliation and operational review should a **capped pilot** permit a small approved universe, capital limit, maximum order size, rate limit, loss limit, and staffed kill switch. Whether any live pilot is appropriate is a separate governance decision, not a course deliverable.

Performance reporting uses the ledger and includes return, volatility, maximum drawdown, expected shortfall, turnover, costs, exposure, rejected and partial orders, stale-data events, reconciliation breaks, and downtime. Compare with passive SPY, the declared cash/STRIPS alternative, equal weight, and no-rebalance variants under the same dates and capital convention. Synthetic success, historical backtest success, and paper success are distinct evidence tiers.

KEY TAKEAWAYS

In this lecture, we converted the course prototype into an auditable launch design with valid preference mathematics, execution accounting, risk controls, and evidence gates.

- **Stay inside the model domain.** Signed SIM beta is not a positive risk scale, signed sentiment is not classical risk aversion, and Cobb–Douglas preferences must remain positive.
- **Fills change the account.** The broker-confirmed ledger, conservation checks, open-order exposure, and reconciliation—not target weights or submitted orders—define holdings, cash, and wealth.
- **Launch is progressive.** Deterministic tests, stress simulation, walk-forward evaluation, paper/shadow operation, and governance precede any tightly capped live pilot.

The course ends with a system that can be questioned, reproduced, monitored, and improved—the essential traits of responsible quantitative engineering.

ADDITIONAL READING

- Marcos López de Prado, *Advances in Financial Machine Learning*, Wiley (2018), chapters on backtest overfitting and feature importance.
- David H. Bailey et al., "The Probability of Backtest Overfitting," *Journal of Computational Finance* 20(4), 39–69 (2017).
- Philip Maymin, "Markets Are Not Continuous," *Journal of Trading* 6(1), 1–8 (2011).

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